Mthunzelwa Makwanyane
20241063
NET731 Practicals – Week3

AZURE CLOUD PROJECT · 3RD YEAR NETWORKING

Week 4 – Identity Management

Microsoft Entra ID: users, groups, and role-based access control

WEEK: 27 Mar – 2 Apr MARKS: 8 PLATFORM: Microsoft Azure / Entra ID SUBMIT: GitHub

TASKS

TASK 01 Plan Your Identity Structure

2 marks

The first part of this deployment is to create a resource group to have a better management of the resources that we will be deploying. Before creating the user, I will create the three groups labelled development, Operations and Finance. Then after creating the groups, I will create the user accounts in the environment and assign them to the specific group they need to be assigned to. When creating the users I will need to integrate the domain name of the organization if they have one. For example, it would be **Firtname.Lastname@NovaTech.onmicrosoft.com** . Since I have a student account, the domain that will automatically be used is my email address on Azure, the outcome will be **Firstname.Lastname@rickd1zeda7gmail.onmicrosoft.com**. Adding the domain name into the user's name is needed for the unique User Principal Name (UPN). This creates a uniqueness across the entire cloud and identifies the specific tenant while showing professional branding.

When creating the three groups in this scenario I will make them security groups in order to be able to manage access control and considering that security groups can include users and devices, not just users. This specific choice of security groups allows the implementation of principle of least privilege, this is the giving a certain amount of access to a user due to the job of the certain user to prevent unwanted access to important data that the user does not need. The membership type is used in this scenario is assigned membership because there is not that many users I need to add whereas if there were 500 user for example, I would have needed to get a Entra ID P2 licence for dynamic membership option. To confirm that the members are added to the group, in the groups tab I would navigate to all groups where all the created groups are shown and enter inside the specific group to see who and how many members are inside said group.

With assigning the built-in roles, for the Finance group I gave them the billing scope reader because al they need to do it be able to analyse data. For the Development team, I granted them the IoT Hub Registry Contributor because they will need access to IoT application and devices. For the operations group, I gave them the reader role because all they need to do is analyse processes in the organisation and there is no need for them to make changes, only suggest them if processes are not in order.

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TASK 02	Create Users and Groups in Entra ID	3 marks

Creating User accounts

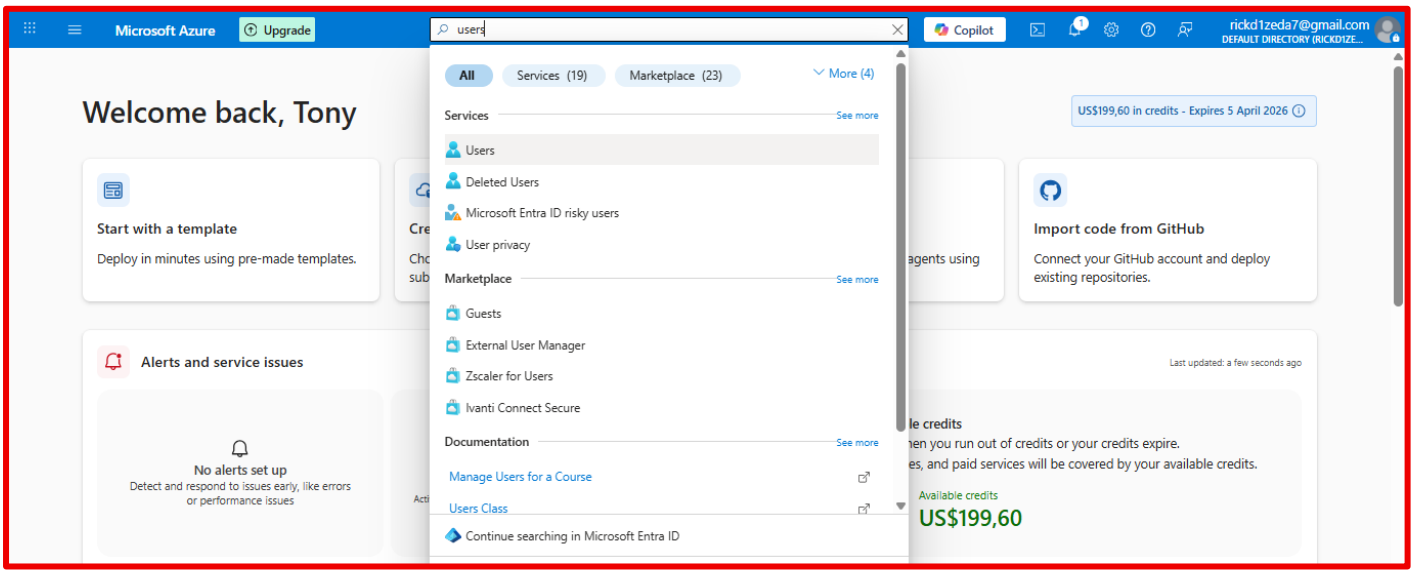

Figure 1: Searching for Users resource

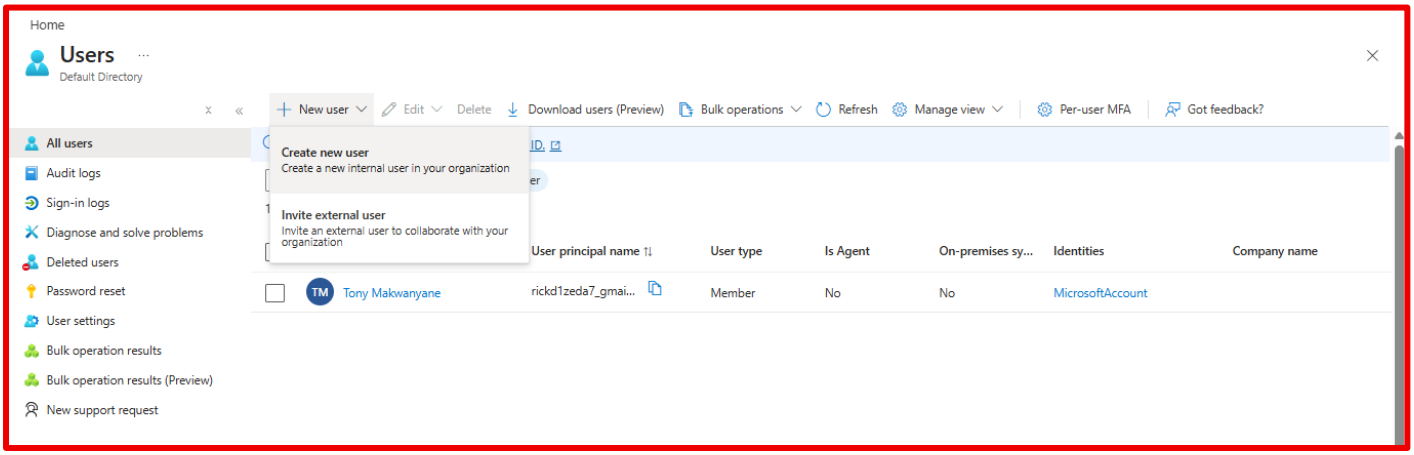

Figure 2: Adding new user

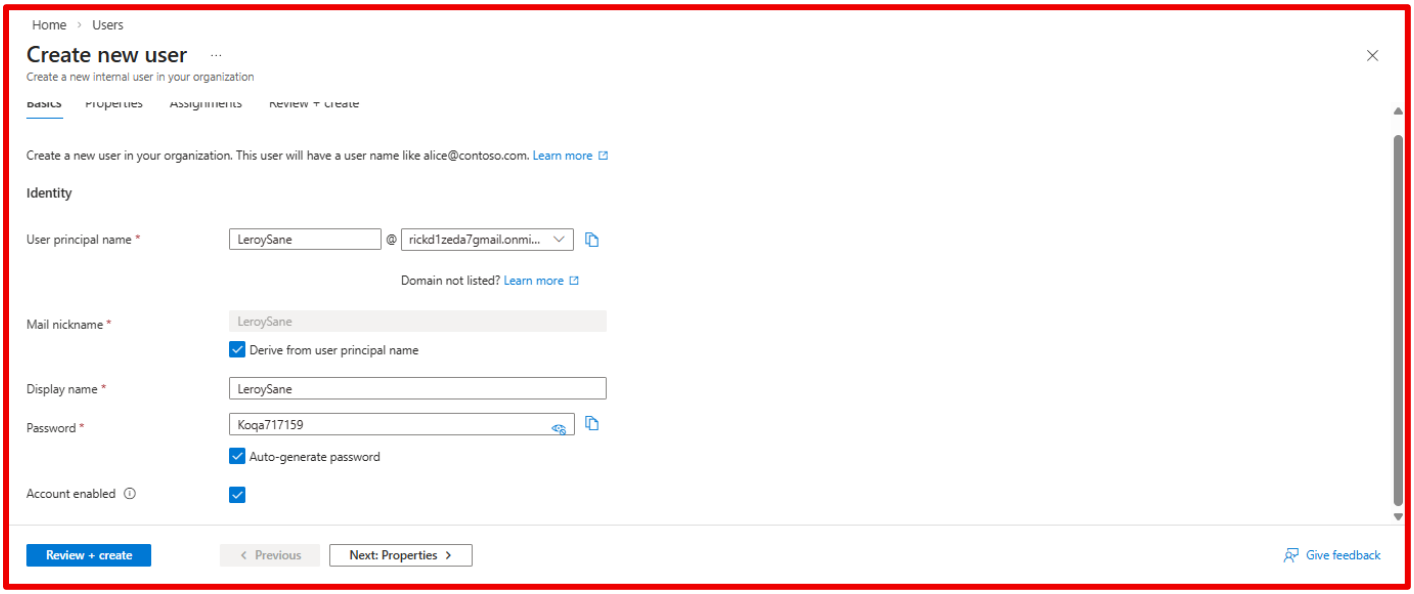

Figure 3: Adding new user info

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Figure 4: Adding new user job info

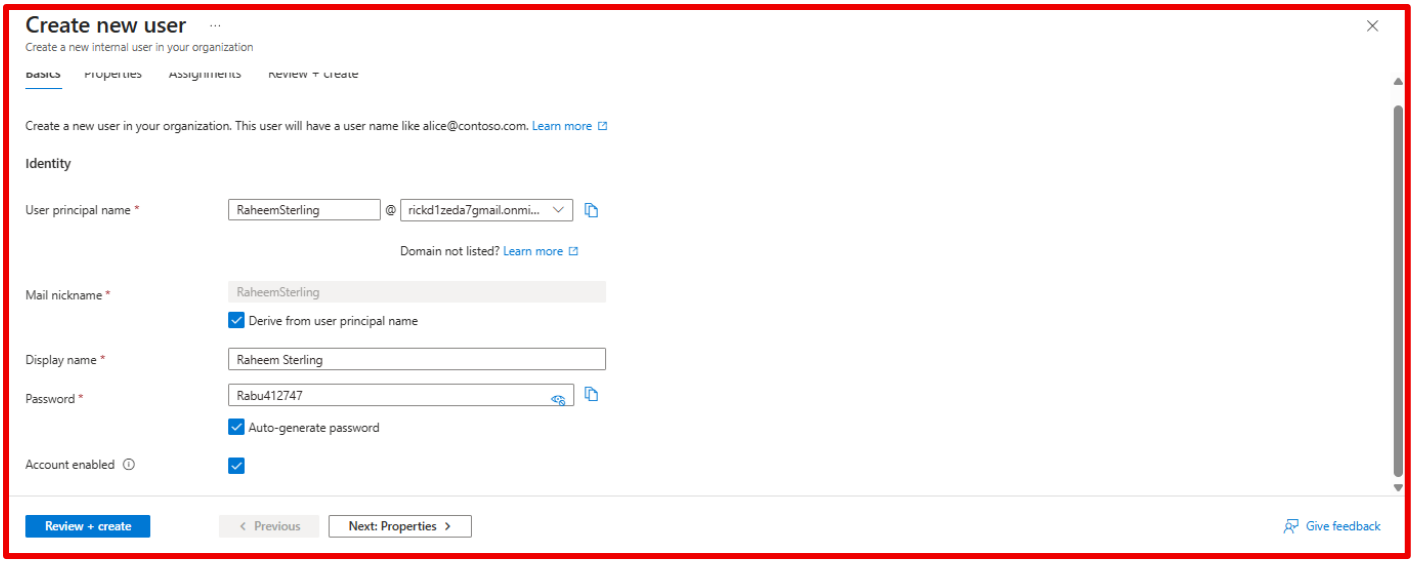

Figure 5: Adding new user info

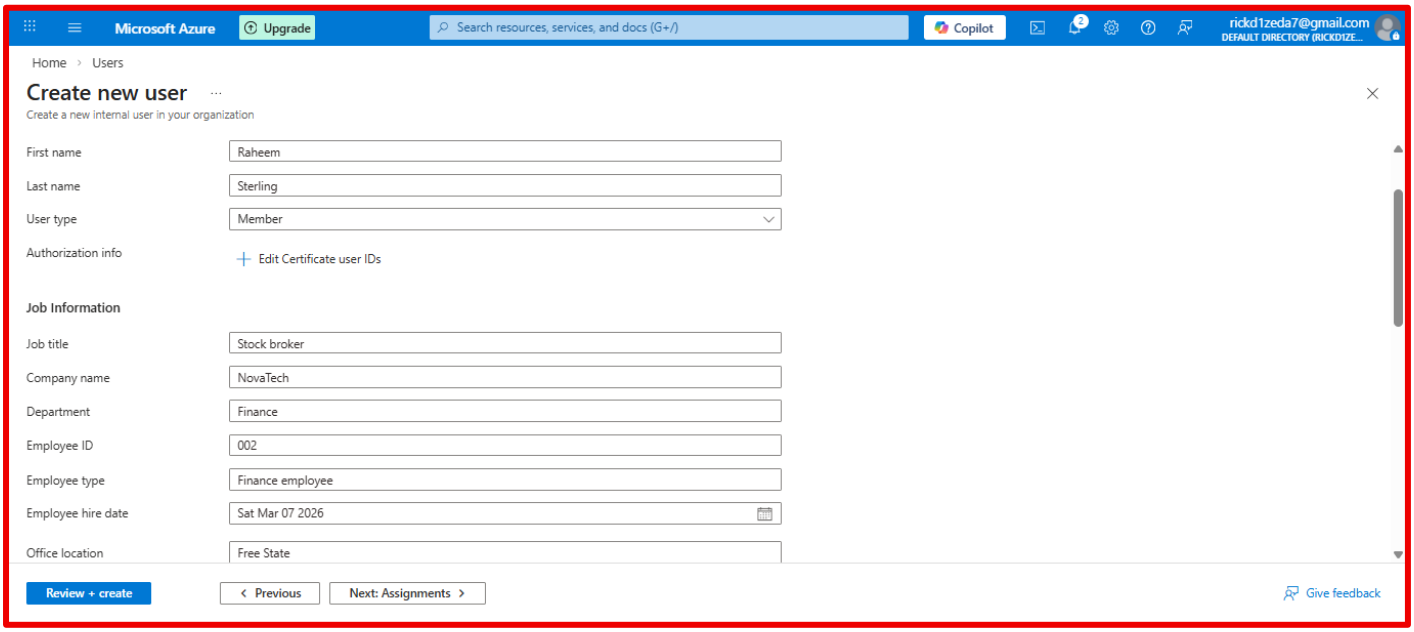

Figure 6: Adding new user job info

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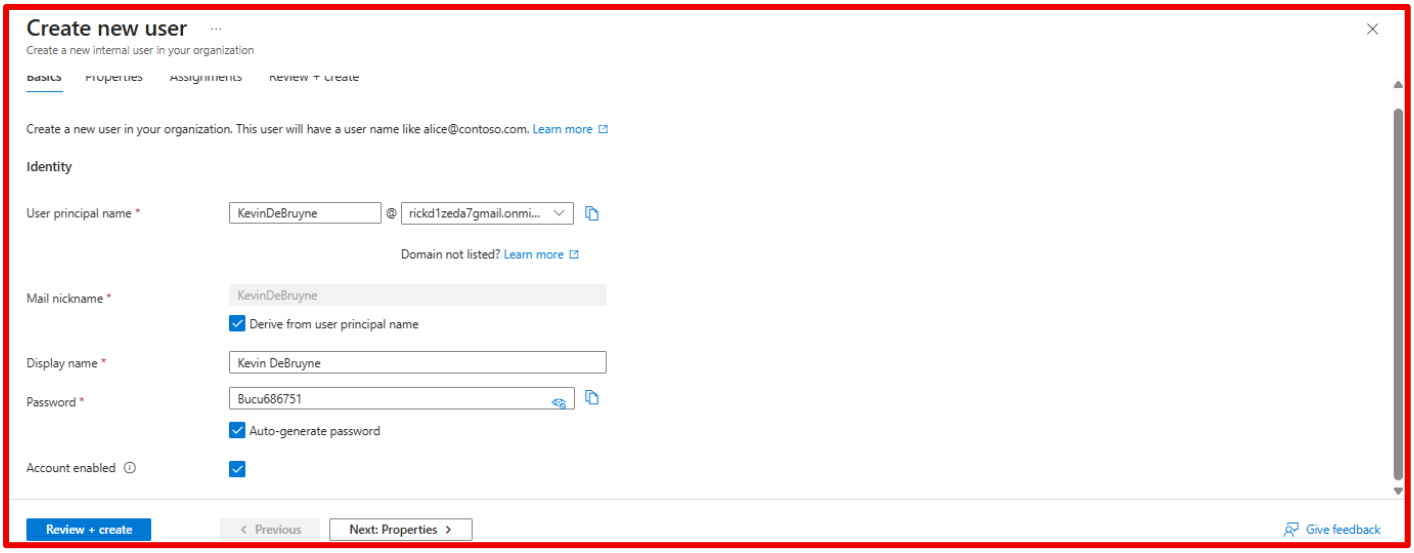

Figure 7: Adding new user info

Figure 8:Adding new user job info

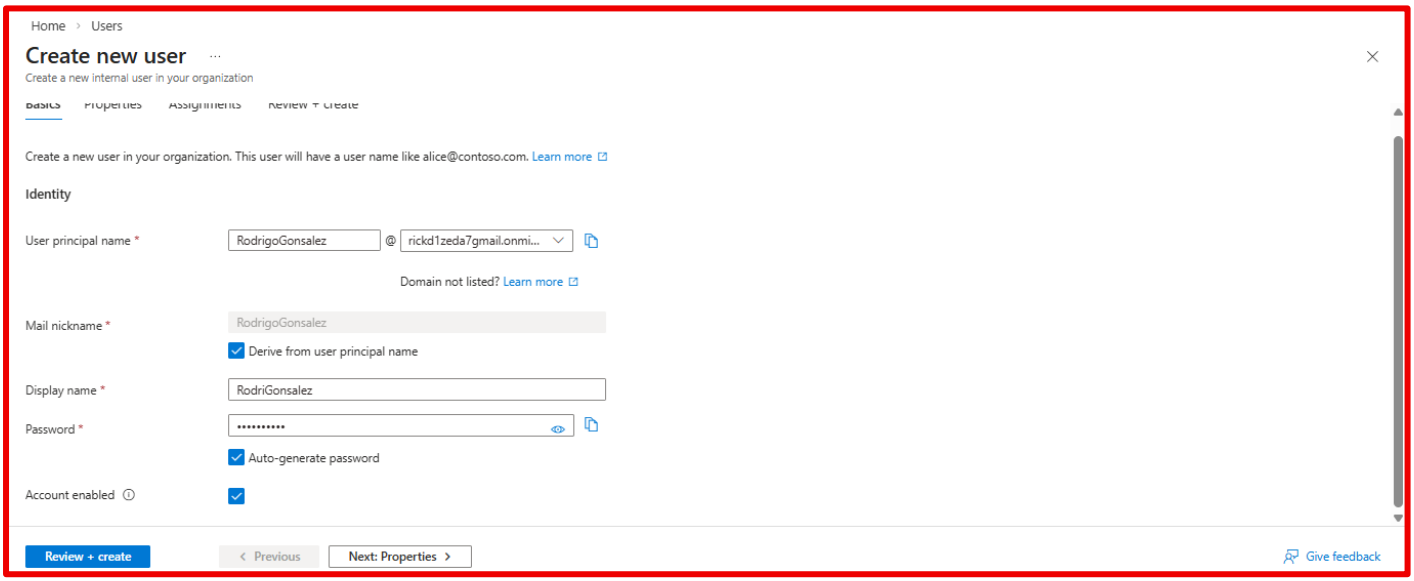

Figure 9: Adding new user info

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Figure 10: Adding new user job info

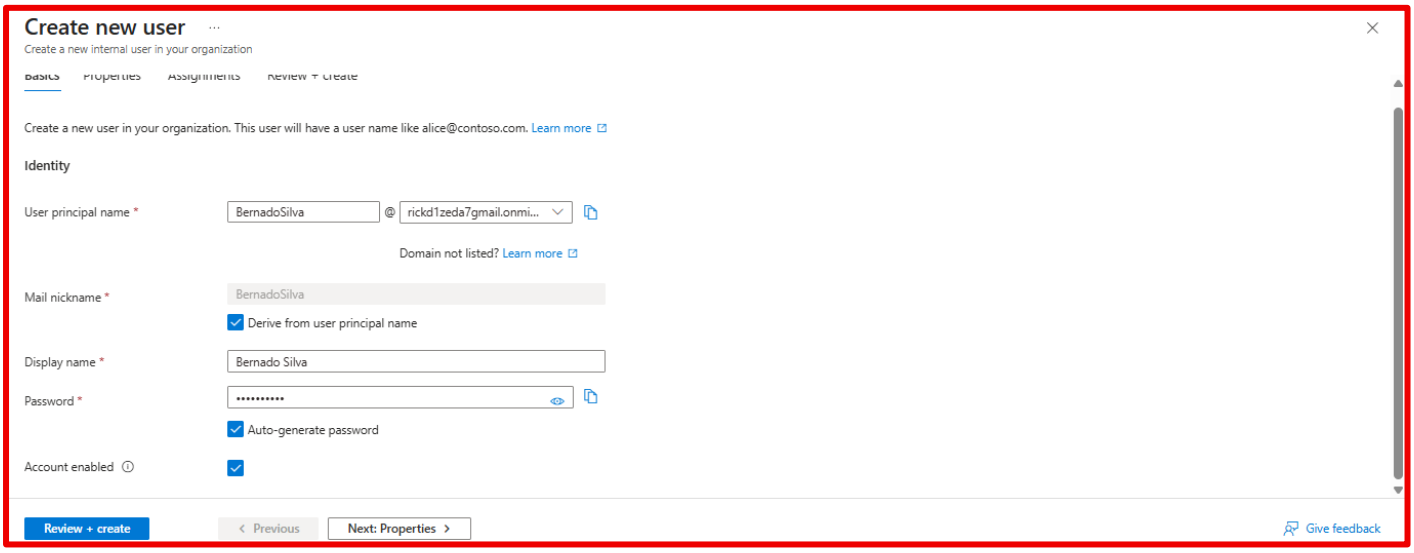

Figure 11: Adding new user info

Figure 12: Adding new user job info

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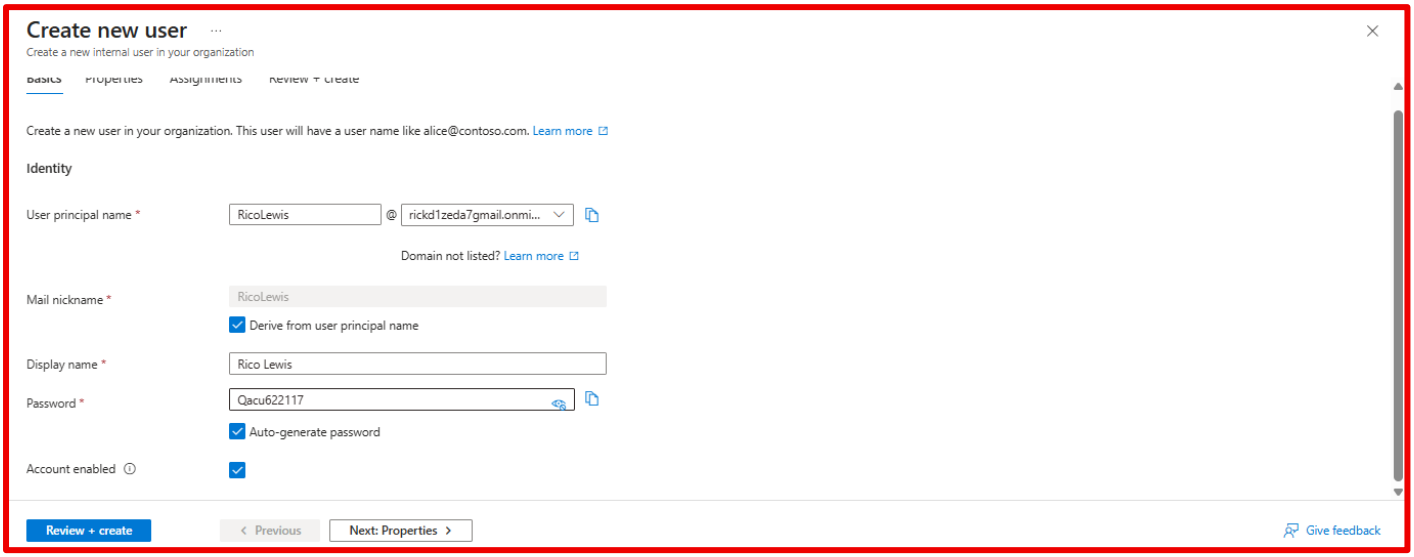

Figure 13: Adding new user info

Figure 14: Adding new user job info

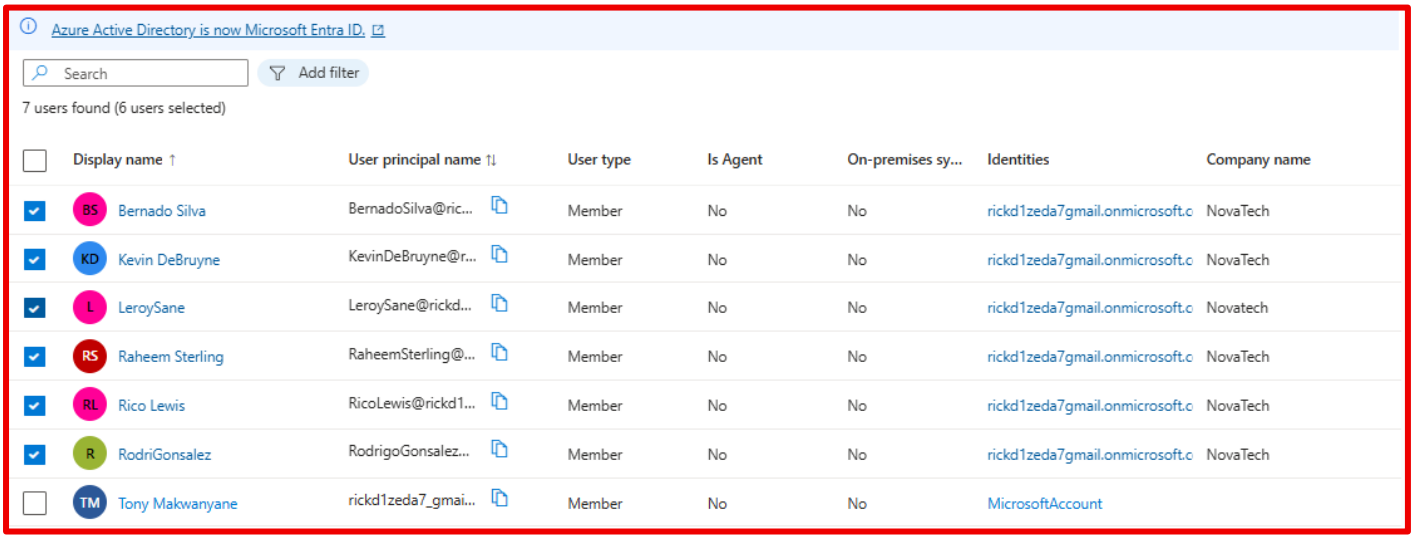

Figure 15: Showing new users created

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Creating security groups

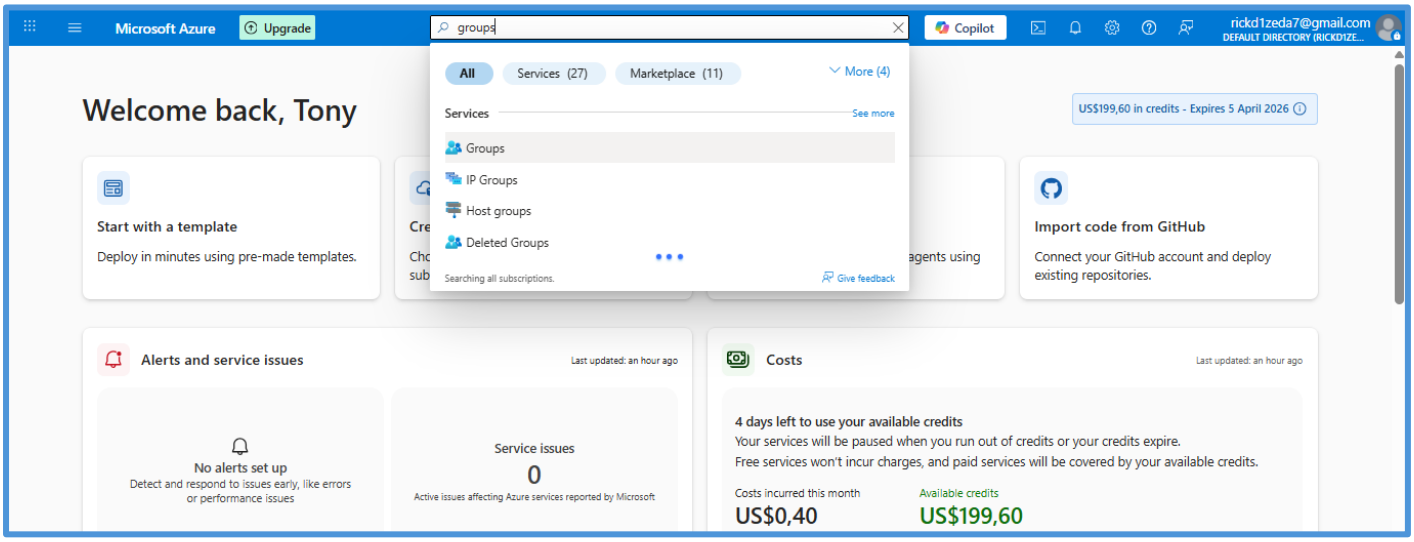

Figure 16: Searching for groups resource

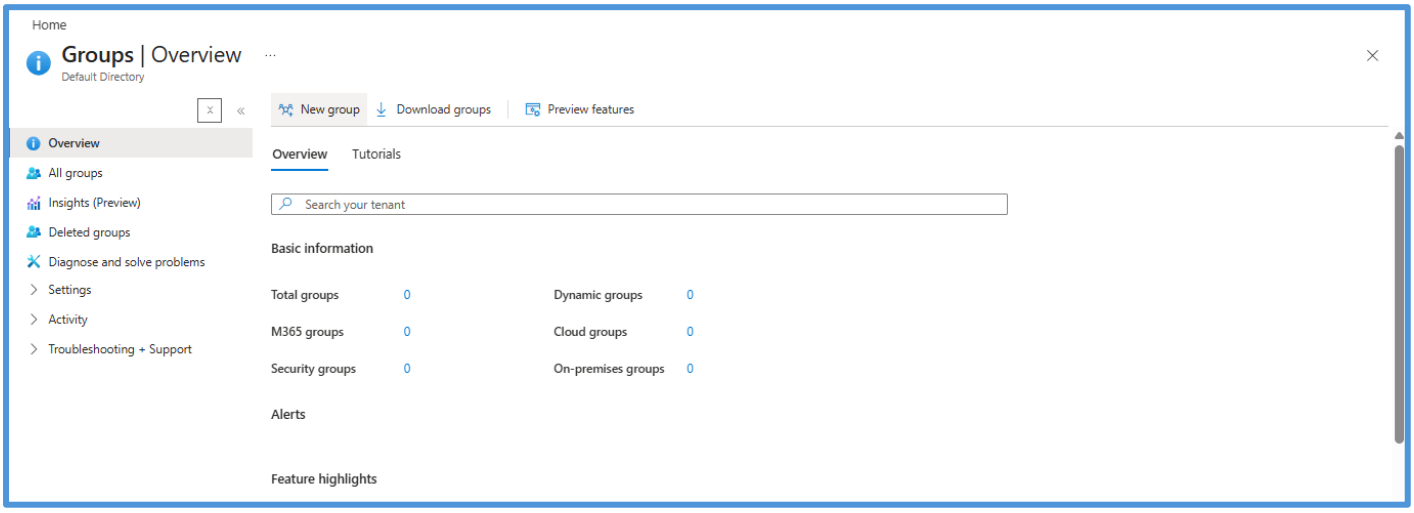

Figure 17: Adding new group

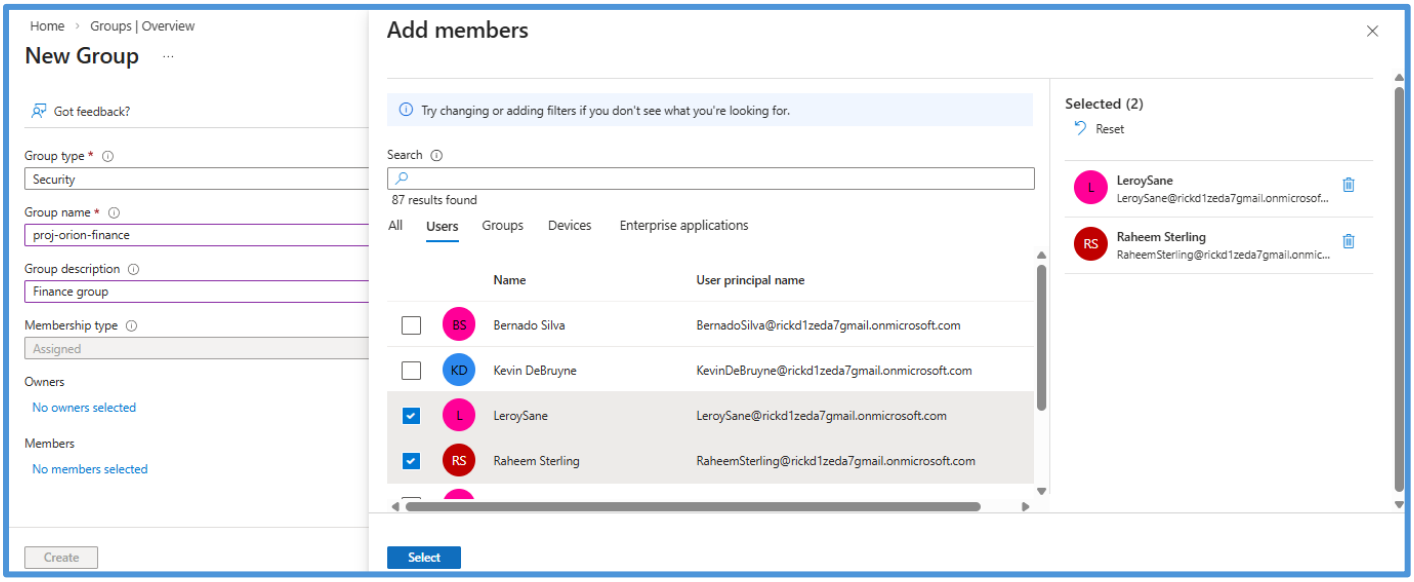

Figure 18: Adding designated users to Finance group

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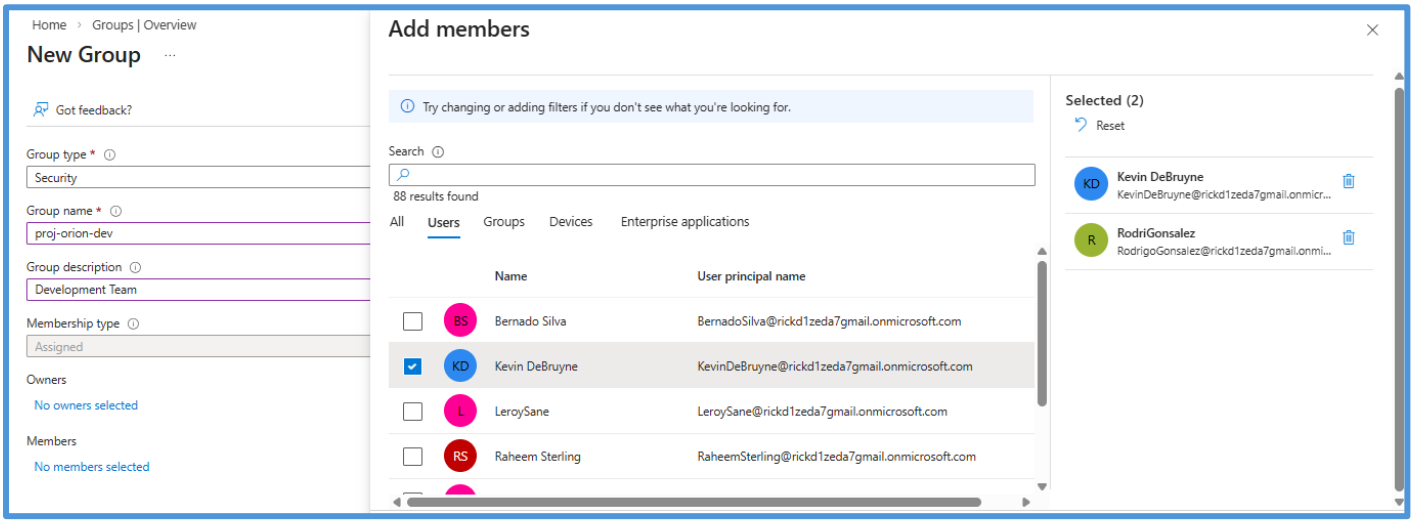

Figure 19: Adding designated users to Development group

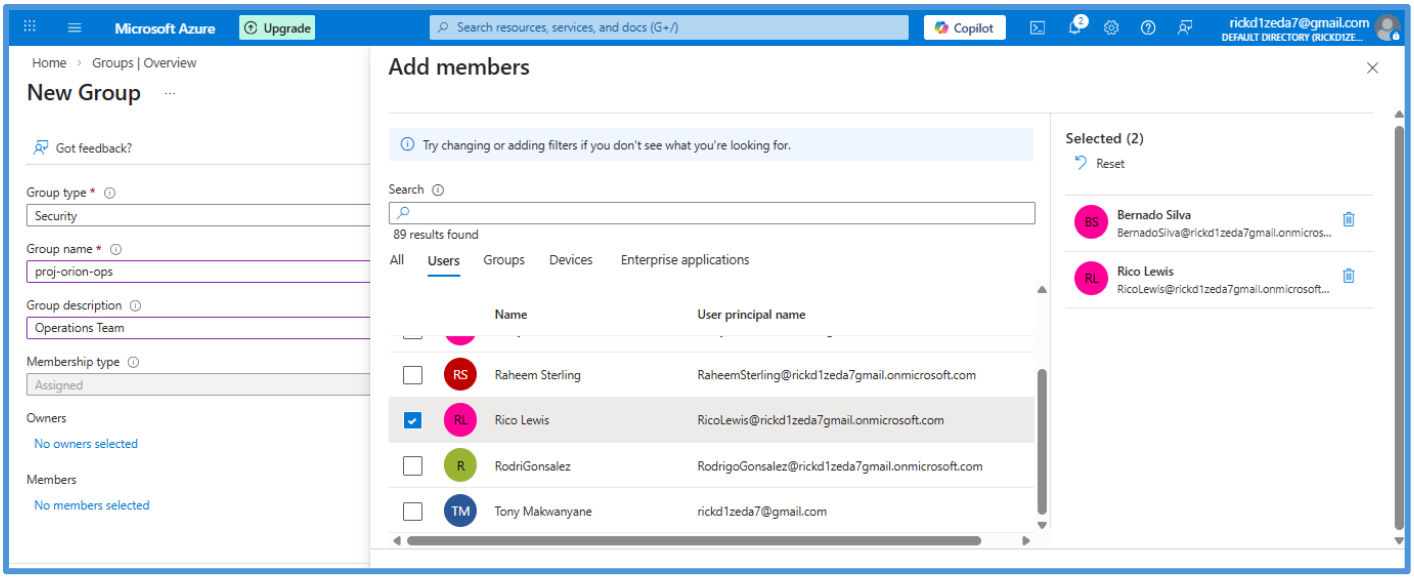

Figure 20: Adding designated users to Operations group

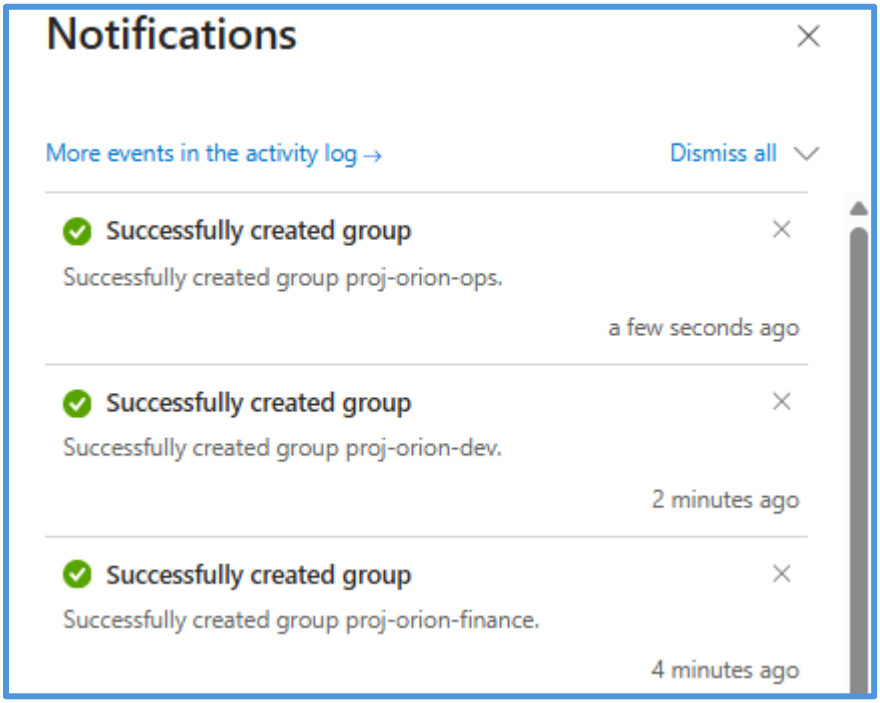

Figure 21: Proof of successful group creations

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Proof of Users in groups

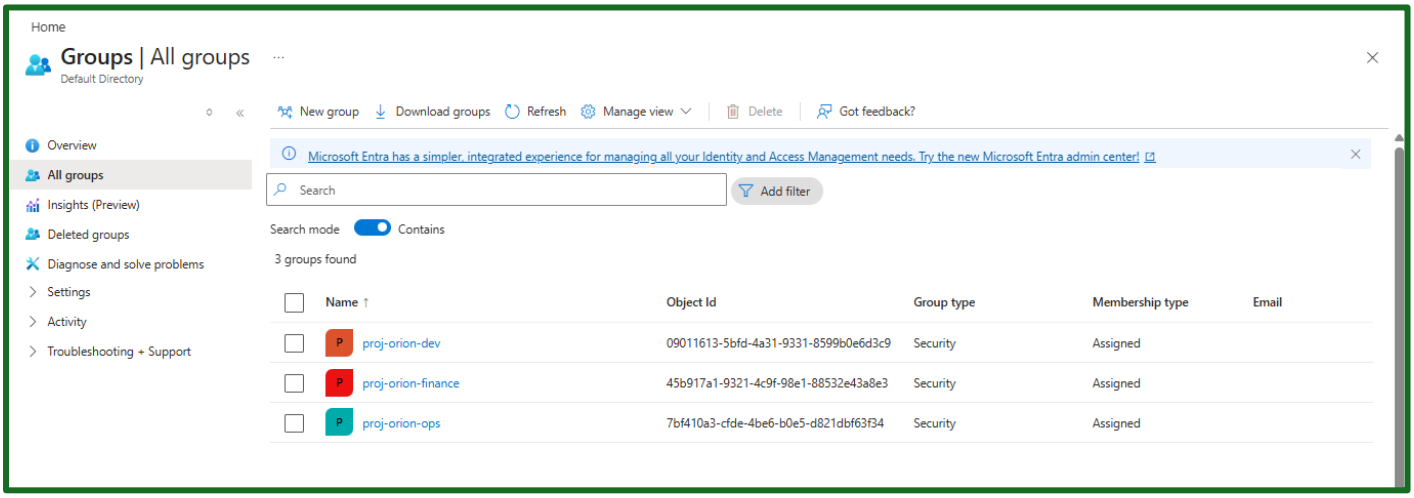

Figure 22: In the All Groups tab to show all the group I have

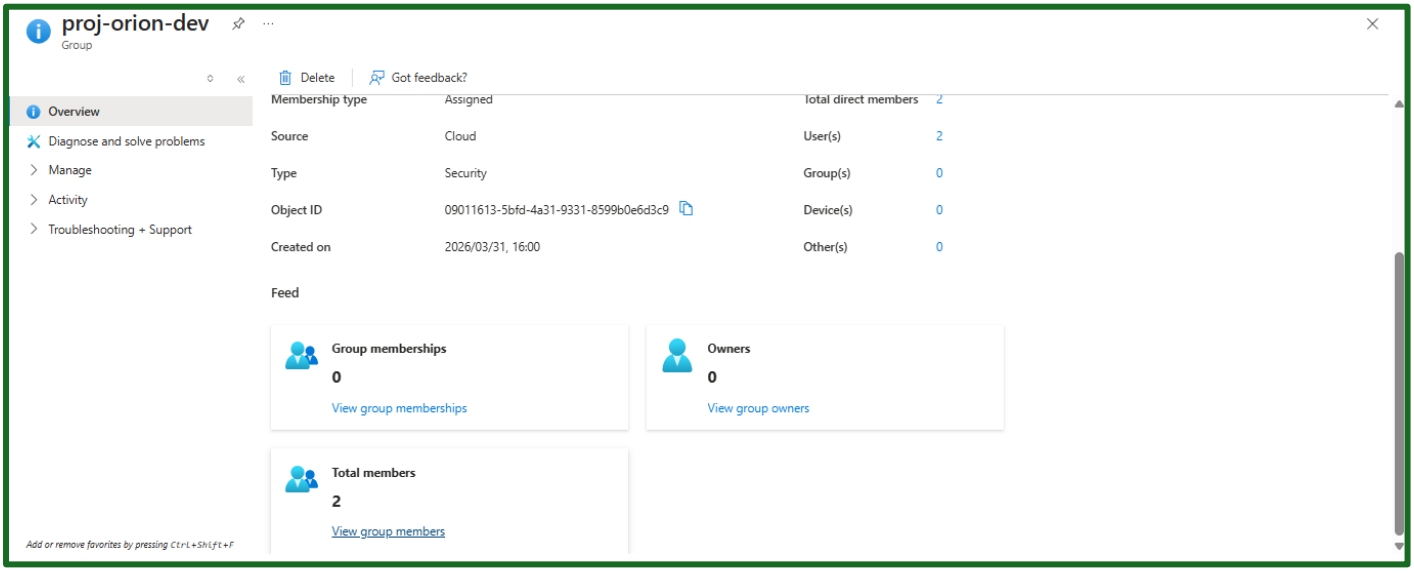

Figure 23: Viewing group statistics

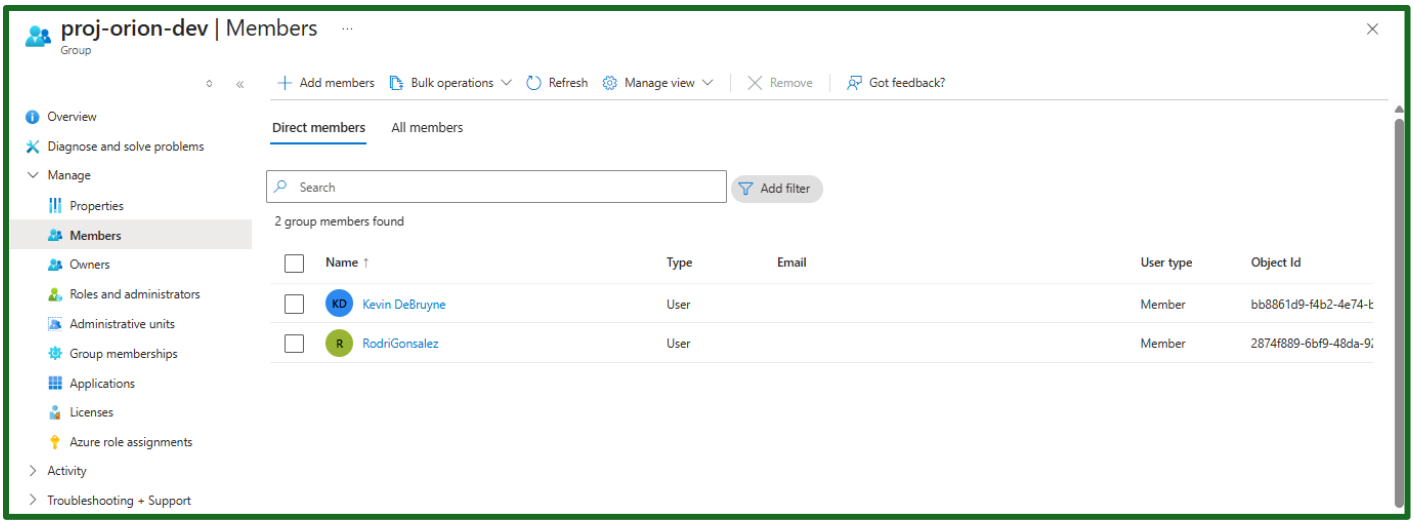

Figure 24: Proof of users in groups

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Creating Built-in roles for groups

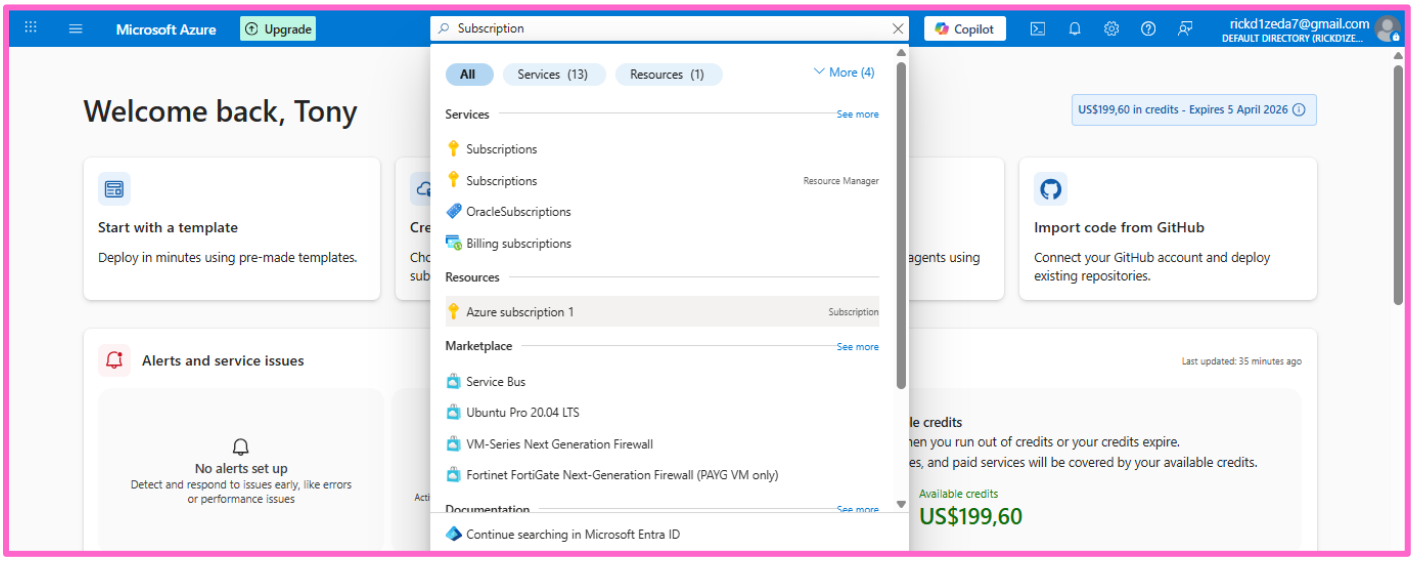

Figure 25: Searching for Subscription resource

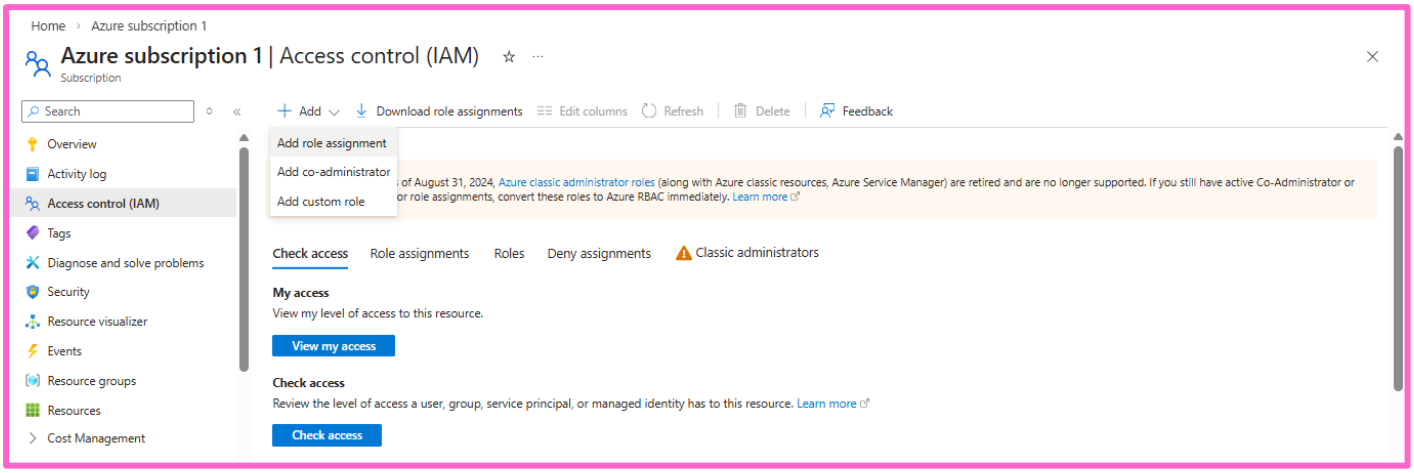

Figure 26: Adding a new role assignment in the IAM tab

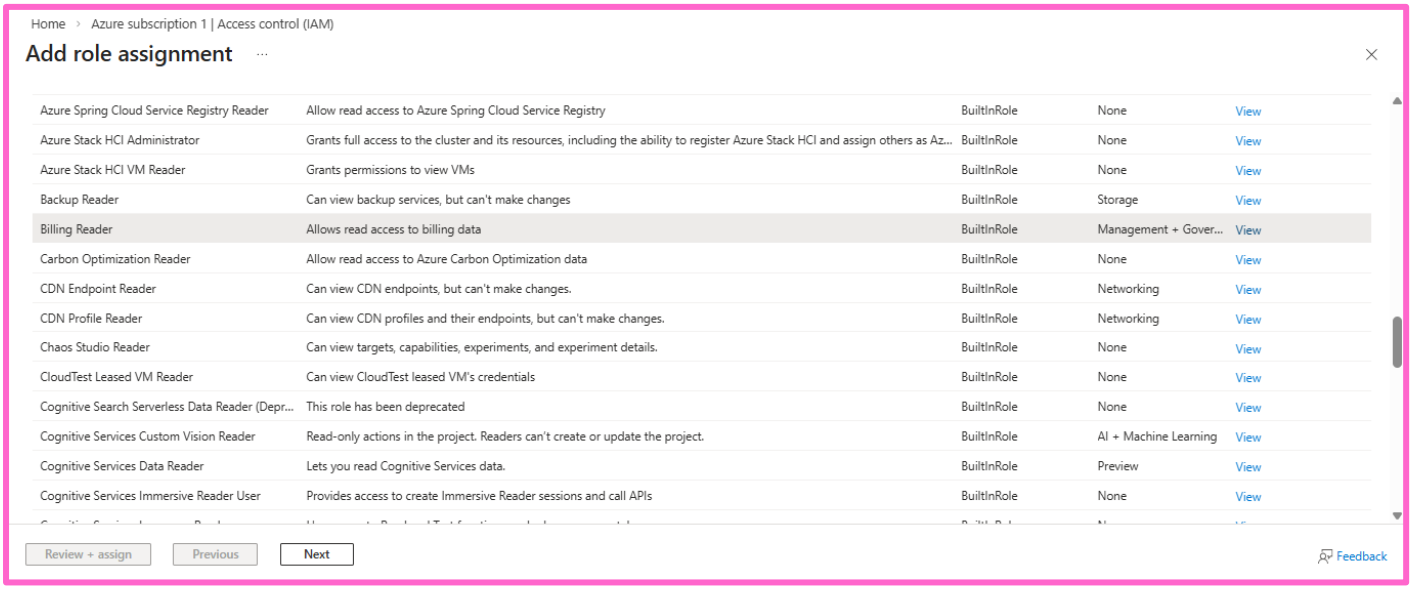

Figure 27: Selecting Finance group role type

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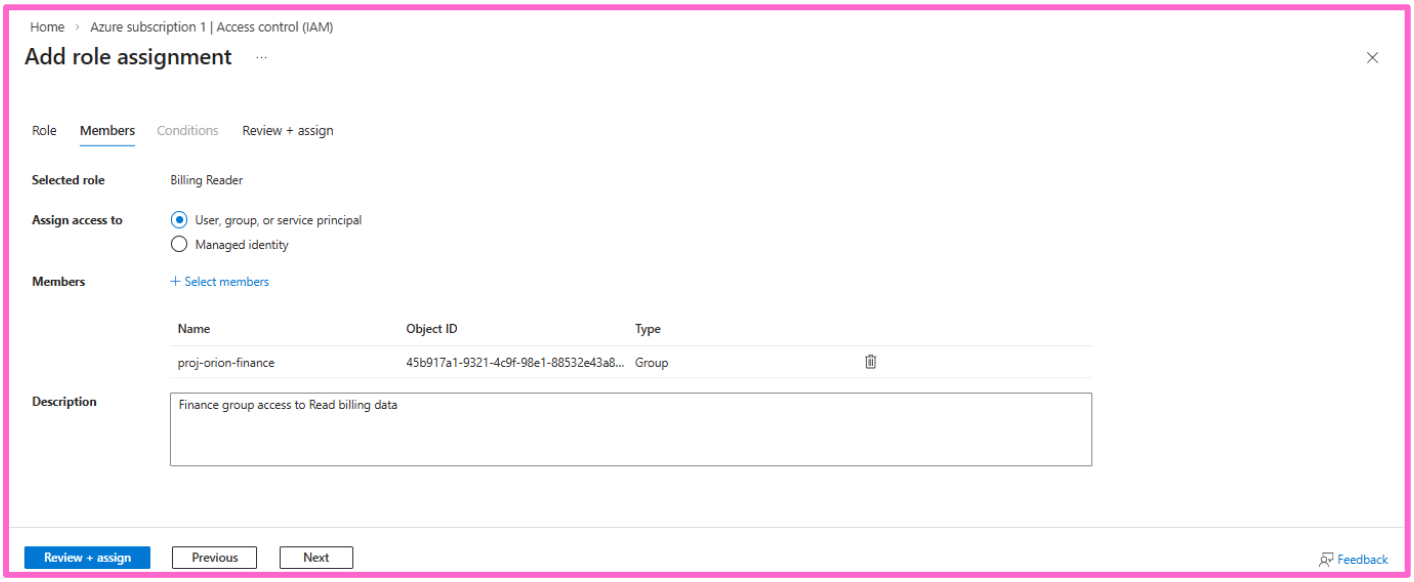

Figure 28: Selecting the Finance group, where the assignment will be applied

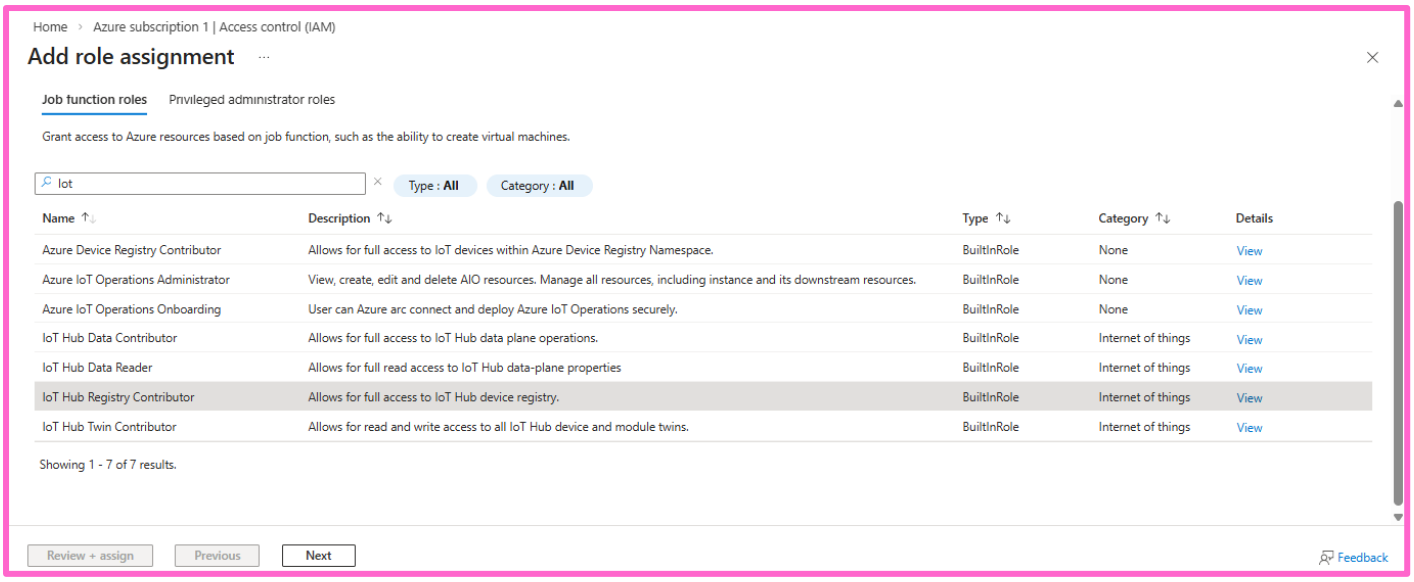

Figure 29: Selecting Development group role type

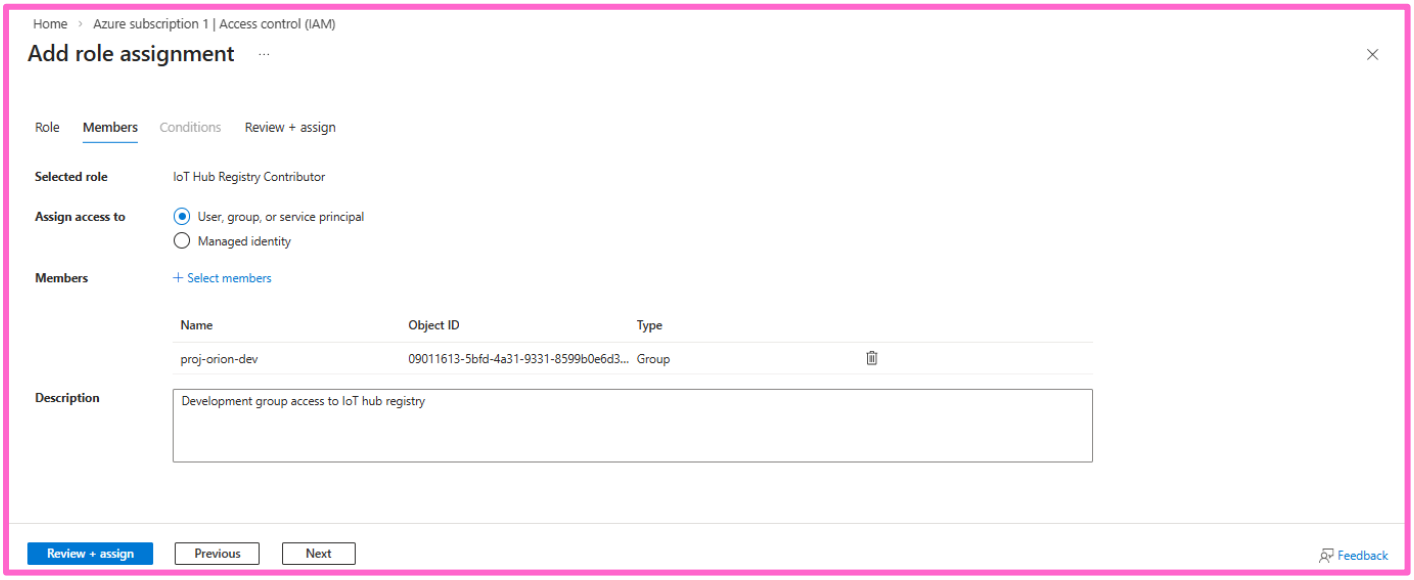

Figure 30: Selecting the Development group, where the assignment will be applied

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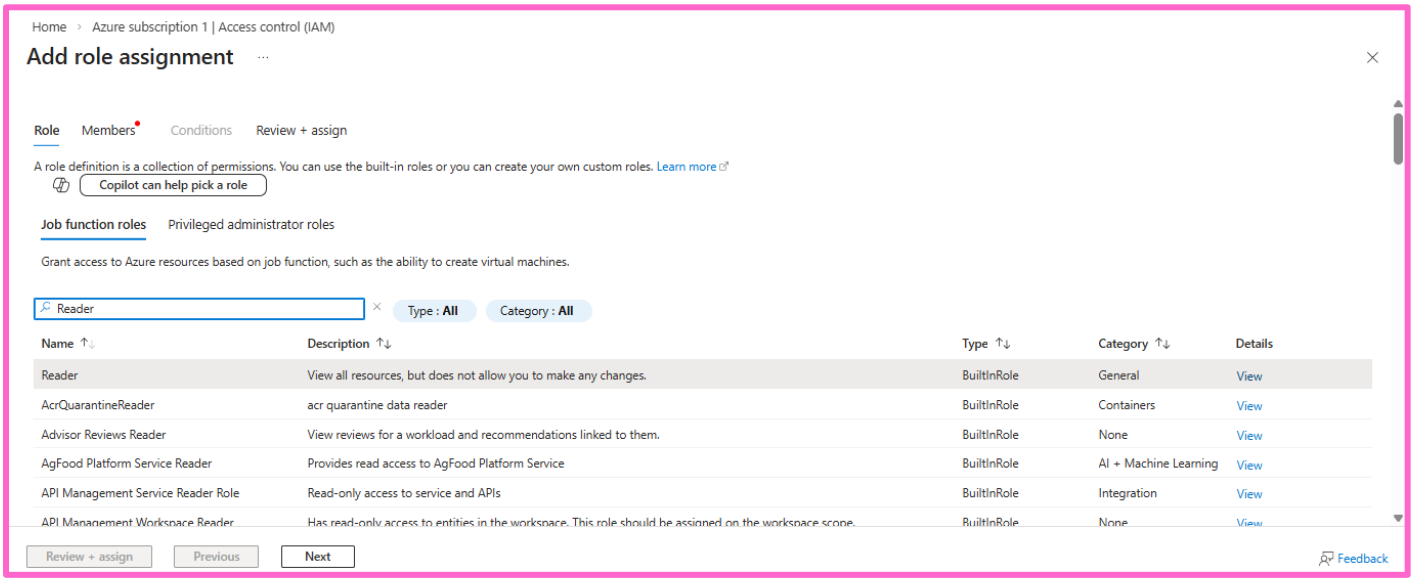

Figure 31: Selecting the role type for the Operations group

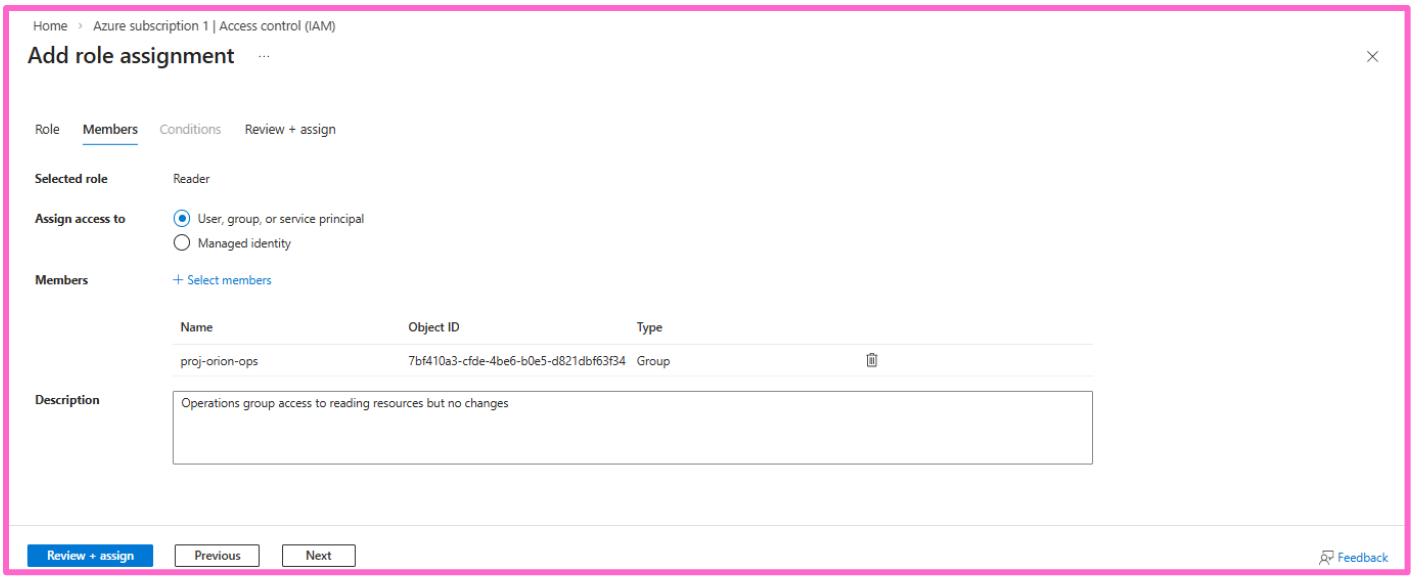

Figure 32: Selecting the Operations resources, where the assignment will be applied

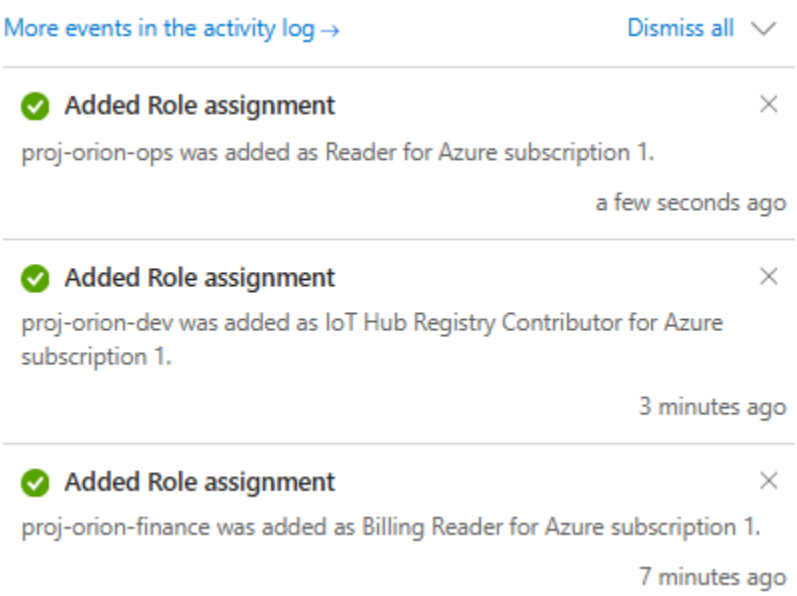

Figure 33: Proof of created role assignments